

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**  
**COURSE CODE: CSA1142**

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### **COURSE OUTCOME COVERED**

**CO5:** Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)  
Assessment Tool Weightage: 35%

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**QUESTIONS: 05**

**Total Marks:35**

Annexure A

Constraint-Based Problem Solving

### **Q3. Scenario: Payroll Management System**

Modules:

Employee Records, Salary Processing, Tax Calculation, Reports

### **QA Constraints:**

- Accurate salary computation
- Data integrity must be ensured
- System must handle bulk processing efficiently

### **Question:**

Use **quality assurance practices** to analyze the system. Identify critical test cases and explain how constraints influence testing priorities and defect prevention. (BL5)

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### **Code of Conduct:**

*I Y. Abhay –(192411219) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

### **Signature of the student:**

**Y. Abhay**

## RUBRICS FOR CALCULATION:

### Constraint-Based Problem Solving Assessment Rubric

| Criteria                | Weightage | Level 4 (Exemplary)                                                            | Level 3 (Proficient)                                  | Level 2 (Developing)                          | Level 1 (Beginning)                      |
|-------------------------|-----------|--------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Problem Understanding   | 25%       | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps           | Partial understanding; misses key constraints | Fails to identify constraints correctly  |
| Constraint Analysis     | 25%       | Analyzes constraints effectively and prioritizes them logically                | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization         | No meaningful analysis or prioritization |
| Solution Development    | 25%       | Develops optimal and feasible solutions satisfying all constraints             | Develops workable solutions with minor limitations    | Solutions partially satisfy constraints       | Solutions do not meet constraints        |
| Application of Concepts | 15%       | Effectively applies appropriate concepts, methods, or tools                    | Applies concepts with minor errors                    | Limited application; requires guidance        | Unable to apply relevant concepts        |
| Justification           | 10%       | Provides strong justification for decisions with logical reasoning             | Provides justification with some clarity              | Weak or unclear justification                 | No justification provided                |

## **Q3. Payroll Management System**

### **Modules**

- Employee Records
- Salary Processing
- Tax Calculation
- Reports

### **QA Constraints**

- Accurate salary computation
- Data integrity must be ensured
- System must handle bulk processing efficiently

## **Quality Assurance (QA) Analysis**

Quality Assurance focuses on preventing defects and ensuring that the Payroll Management System meets functional and performance requirements.

### **QA Practices**

1. Requirement Review
  - Verify salary rules, tax regulations, and report requirements.
2. Test Planning
  - Define test cases for salary calculation, tax deduction, and report generation.
3. Code Review
  - Check formulas used in salary and tax calculations.
4. Functional Testing
  - Validate all payroll operations.
5. Performance Testing
  - Test processing of large numbers of employee records.
6. Data Validation Testing
  - Ensure employee information remains accurate and consistent.

## Critical Test Cases

### 1. Employee Records Module

| Test Case ID | Test Scenario           | Expected Result                     |
|--------------|-------------------------|-------------------------------------|
| TC01         | Add new employee        | Employee record stored successfully |
| TC02         | Duplicate Employee ID   | Error message displayed             |
| TC03         | Update employee details | Record updated correctly            |
| TC04         | Delete employee record  | Record removed successfully         |

### 2. Salary Processing Module

| Test Case ID | Test Scenario                    | Expected Result                 |
|--------------|----------------------------------|---------------------------------|
| TC05         | Calculate salary with allowances | Correct salary generated        |
| TC06         | Employee with overtime           | Overtime added correctly        |
| TC07         | Employee with deductions         | Net salary calculated correctly |
| TC08         | Zero basic salary                | Validation error displayed      |

### 3. Tax Calculation Module

| Test Case ID | Test Scenario                      | Expected Result      |
|--------------|------------------------------------|----------------------|
| TC09         | Calculate tax for taxable employee | Correct tax deducted |
| TC10         | Employee below tax slab            | No tax deduction     |
| TC11         | Incorrect tax rate entry           | Validation error     |

### 4. Reports Module

| Test Case ID | Test Scenario                   | Expected Result             |
|--------------|---------------------------------|-----------------------------|
| TC12         | Generate monthly payroll report | Accurate report generated   |
| TC13         | Generate report with no records | "No Data Available" message |
| TC14         | Export payroll report           | Successful export           |

### 5. Bulk Processing Test Cases

| Test Case ID | Test Scenario                       | Expected Result                  |
|--------------|-------------------------------------|----------------------------------|
| TC15         | Process salary for 10,000 employees | Completed within acceptable time |
| TC16         | Generate reports for large dataset  | System remains responsive        |

# Influence of Constraints on Testing Priorities

## 1. Accurate Salary Computation

**Priority: Very High**

Because salary errors directly affect employees and organization finances.

### Testing Focus

- Gross salary calculation
- Overtime calculation
- Deductions
- Net salary calculation

### Defect Prevention

- Formula verification
- Boundary value testing
- Automated calculation checks

## 2. Data Integrity

**Priority: High**

Employee records must remain accurate and secure.

### Testing Focus

- Duplicate employee IDs
- Data consistency
- Database validation
- Record update accuracy

### Defect Prevention

Input Validation



Database Constraints



Data Consistency Checks



### 3. Bulk Processing Efficiency

**Priority: High**

The system should process thousands of employees without delays.

**Testing Focus**

- Load Testing
- Stress Testing
- Performance Testing

**Defect Prevention**

- Database optimization
- Efficient queries
- Resource monitoring

### Testing Priority Matrix

| Constraint      | Priority  | Testing Type                  |
|-----------------|-----------|-------------------------------|
| Salary Accuracy | Very High | Functional Testing            |
| Data Integrity  | High      | Database & Validation Testing |
| Bulk Processing | High      | Performance Testing           |
| Report Accuracy | Medium    | Functional Testing            |
| User Interface  | Medium    | Usability Testing             |

### Defect Prevention Techniques

**Requirement Validation**

- Verify payroll rules before development.

**Peer Reviews**

- Review salary and tax calculation logic.

## **Automated Testing**

- Run repetitive payroll test cases automatically.

## **Regression Testing**

- Ensure new changes do not affect existing payroll functions.

## **Performance Monitoring**

- Detect bottlenecks during bulk processing.

## **Conclusion**

Quality Assurance plays a critical role in the Payroll Management System by ensuring accurate salary computation, maintaining data integrity, and supporting efficient bulk processing. Critical test cases focus on employee records, salary calculations, tax deductions, and report generation. The given constraints determine testing priorities, while QA practices such as reviews, validation, automation, and performance testing help prevent defects and improve overall system reliability and quality.